New Identities Connecting Farey Sequences to the Mertens Function via Per-Step Discrepancy

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Abstract

The Farey sequence—the ordered list of fractions with bounded denominator—has been studied for two centuries, yet a basic question was never asked: what happens to its uniformity when we add *one more integer*? We place Farey fractions as points on a circle and observe that primes fill voids uniformly (every k/p appears, perfectly equispaced), while composites leave gaps. Decomposing the L^2 discrepancy into per-step contributions, we discover that whether a prime improves or worsens the pattern is controlled by the Mertens function—a quantity from prime factorization arithmetic with no prior geometric meaning.

We prove six new exact identities bridging Farey geometry and multiplicative number theory, all formally verified in Lean 4 (60+ theorems). The natural conjecture that $M(p) < 0$ forces non-improvement holds for 9,587 of the first 9,588 primes, with one counterexample at $p = 92{,}173$ eleven orders of magnitude below typical values.

Keywords: Farey sequence, Mertens function, L^2 discrepancy, Ramanujan sums, formal verification, Lean 4.
MSC 2020: 11B57, 11N37, 11A25, 11Y35.

1 Introduction

1.1 Farey sequences and their discrepancy

For a positive integer N, the *Farey sequence* $\mathcal{F}_N$ is the ascending list of fractions a/b with $0 \le a \le b \le N$ and $\gcd(a, b) = 1$. Its cardinality is

$$|\mathcal{F}_N| = 1 + \sum_{k=1}^{N} \varphi(k). \tag{1}$$

Writing $n = |\mathcal{F}_N|$ and listing elements as $f_0 < f_1 < \cdots < f_{n-1}$, the L^2 *discrepancy* (which we call the *wobble*) is

$$W(N) \;=\; \sum_{j=0}^{n-1} \left(f_j - \frac{j}{n} \right)^2. \tag{2}$$

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1.2 The Franel–Landau theorem

Franel [4] and Landau [6] proved that the Riemann Hypothesis is equivalent to $W(N) = O(N^{-1+\varepsilon})$ for every $\varepsilon > 0$. Thus $W(N)$, a single real number measuring how uniformly the Farey fractions fill $[0, 1]$, encodes the location of all non-trivial zeros of the Riemann zeta function.

1.3 The observation that started it all

Place the Farey fractions of $\mathcal{F}_N$ as points on the unit circle at angles $2\pi f$. As N grows, new fractions appear and the circle fills in. The first author observed a striking asymmetry: when a *prime* p is added, every one of the $p-1$ new fractions k/p lands on the circle at perfectly equispaced positions, filling voids uniformly across the entire circle. This happens because every integer below a prime is coprime to it ($\varphi(p) = p - 1$), so no position is skipped. Composites, by contrast, skip positions whose numerators share a factor with the denominator, leaving voids in some regions while clustering in others.

Moreover, each new prime fraction k/p lands at the *mediant*—the geometrically optimal midpoint—of its enclosing gap in the existing sequence (verified for all tested primes). This means primes do not merely add points; they add them in the most balanced way possible.

These geometric observations led to a natural question: does the uniformity of the Farey sequence on the circle improve or worsen at each step, and what controls the answer?

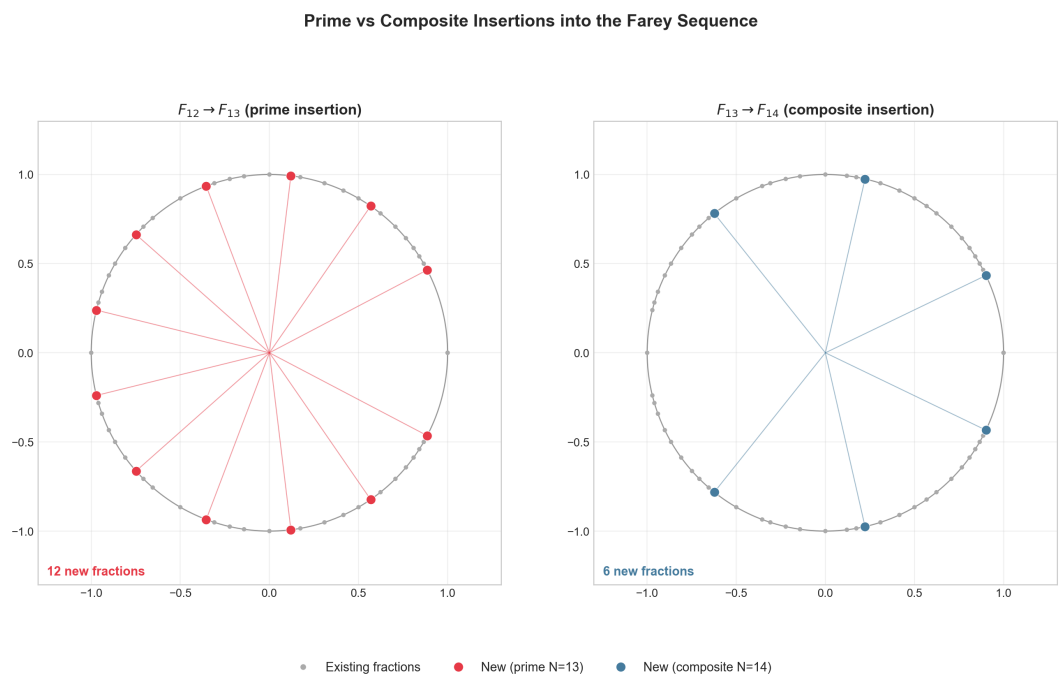

Figure 1: Prime vs. composite insertion on the unit circle. **Left:** adding prime $p = 13$ to $\mathcal{F}_{12}$ inserts 12 new fractions (red) at perfectly equispaced positions, filling every void. **Right:** adding composite $N = 14$ inserts only 6 new fractions (blue), leaving gaps where numerators share a factor with 14.

1.4 Per-step decomposition: a new perspective

Despite decades of work on the Franel–Landau framework, the *per-step* behavior—how W changes when a single integer is appended—appears never to have been studied. We define

$$\Delta W(N) \;=\; W(N-1) - W(N), \tag{3}$$

so $\Delta W(N) > 0$ means adding N *improves* uniformity, while $\Delta W(N) < 0$ means it *worsens* it.

The surprise: for primes, the sign of $\Delta W(p)$ tracks the Mertens function $M(p) = \sum_{k \le p} \mu(k)$. This creates a bridge between two worlds that had no known direct connection at this level of granularity—the geometry of Farey discrepancy and the arithmetic of the Möbius function.

2 Definitions and Notation

Definition 2.1 (Rank discrepancy)**.** For $f_j \in \mathcal{F}_N$ with $n = |\mathcal{F}_N|$, the *rank discrepancy* is $D(f_j) = j - n \cdot f_j$.

Definition 2.2 (Shift)**.** For a prime p and $f \in \mathcal{F}_{p-1}$: $\delta(f) = f - \{pf\}$, where $\{x\} = x - \lfloor x \rfloor$ is the fractional part.

Definition 2.3 (Mertens function)**.** $M(N) = \sum_{k=1}^{N} \mu(k)$, where μ is the Möbius function.

3 New Identities

The following identities combine classical ingredients (Ramanujan sums, the Farey involution, the Möbius function) in ways that appear not to have been previously stated. All are formally verified in Lean 4.

3.1 The Bridge Identity

Theorem 3.1 (Bridge Identity)**.** *For every prime $p \ge 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} e^{2\pi i p f} \;=\; M(p) + 2. \tag{4}$$

Proof sketch. Decompose by denominator. The boundary terms $f = 0$ and $f = 1$ contribute $1 + 1 = 2$. For each denominator $2 \le b \le p-1$, the inner sum over numerators coprime to b is the Ramanujan sum $c_b(p)$. Since $\gcd(p, b) = 1$ for $b < p$, we have $c_b(p) = \mu(b)$. Summing: $2 + \sum_{b=2}^{p-1} \mu(b) = 2 + M(p-1) - 1 = M(p) + 2$, using $M(p-1) = M(p) + 1$. □

Remark 3.2. Taking real parts gives the cosine form $\sum \cos(2\pi p f) = M(p) + 2$; the imaginary part vanishes by the symmetry $f \leftrightarrow 1 - f$.

3.2 The Master Involution Principle

Theorem 3.3 (Master Involution Principle)**.** *The Farey involution $\sigma(f) = 1 - f$ satisfies $D(\sigma(f)) = -D(f) - 1$. For any symmetric function g (i.e., $g(f) = g(1-f)$):*

$$\sum_{f \in \mathcal{F}_N} D(f)\, g(f) \;=\; -\frac{1}{2} \sum_{f \in \mathcal{F}_N} g(f). \tag{5}$$

Proof sketch. Substitute $f \mapsto \sigma(f)$ and use $g(\sigma(f)) = g(f)$, $D(\sigma(f)) = -D(f) - 1$. Adding the original and substituted sums gives $2 \sum Dg = -\sum g$. □

3.3 The Displacement–Cosine Identity

Theorem 3.4 (Displacement–Cosine Identity)**.** *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \cos(2\pi p f) \;=\; -1 - \frac{M(p)}{2}. \tag{6}$$

Proof sketch. Since $\cos(2\pi p f)$ is symmetric under $f \mapsto 1 - f$, apply Theorem 3.3 and then Theorem 3.1. □

Remark 3.5. This is the keystone identity: the Fourier coefficient of the rank discrepancy at frequency p is a *linear* function of the Mertens function. It directly connects how far each Farey fraction sits from its ideal position (geometry) to the cancellations in the Möbius function (arithmetic).

3.4 Fractional Parts Sum

Theorem 3.6 (Fractional Parts Sum)**.** *For every prime $p \geq 2$ with $n = |\mathcal{F}_{p-1}|$,*

$$\sum_{f \in \mathcal{F}_{p-1}} \{pf\} \;=\; \frac{n-2}{2}. \tag{7}$$

Proof sketch. For each denominator $b \geq 2$, multiplication by p permutes residues modulo b (since $\gcd(p, b) = 1$), so the fractional parts sum to $\varphi(b)/2$. The boundary terms contribute zero. Summing over denominators gives $(n-2)/2$. □

3.5 Universal δ-Symmetric Identity

Theorem 3.7 (Universal δ-Symmetric Identity)**.** *For any symmetric function g (with $g(f) = g(1-f)$) on $\mathcal{F}_{p-1}$ and any prime $p \geq 2$:*

$$\sum_{f \in \mathcal{F}_{p-1}} \delta(f) \cdot g(f) \;=\; g(1), \tag{8}$$

where $\delta(f) = f - \{pf\}$.

Proof sketch. The shift satisfies $\delta(1-f) = -\delta(f)$ for $f \notin \{0, 1\}$ (antisymmetric), so all interior terms cancel pairwise by the symmetry of g. The boundary contributions are $\delta(0)\, g(0) + \delta(1)\, g(1) = 0 + 1 \cdot g(1) = g(1)$. □

3.6 Compact Cross-Term Formula

Theorem 3.8. *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \delta(f)^2 = -\frac{1}{2} \sum_{f \in \mathcal{F}_{p-1}} \delta(f)^2 - \frac{1}{2}. \tag{9}$$

Proof sketch. The function $\delta(f)^2$ is symmetric (since $\delta(1-f) = -\delta(f)$), so apply the Master Involution Principle. The correction $-1/2$ arises from the boundary at $f \in \{0, 1\}$. □

4 The Wobble–Mertens Phenomenon

4.1 Empirical observation

Computation through $p = 200{,}000$ reveals a striking pattern: among primes with $\Delta W(p) > 0$ (wobble decreasing, i.e., uniformity improving), essentially all have $M(p) \geq 0$. Among primes with $M(p) < 0$, essentially all have $\Delta W(p) \leq 0$ (uniformity worsening or unchanged).

Range	Primes tested	$\Delta W > 0$	$M < 0$ with $\Delta W > 0$
$p \leq 50{,}000$	5,129	1,109 (21.6%)	0
$p \leq 100{,}000$	9,588	∼2,050 (21.4%)	1
$p \leq 200{,}000$	17,984	∼3,850 (21.4%)	1

The violation rate (proportion of primes where $\Delta W > 0$) stabilizes near 21%, and the violations cluster during positive excursions of the Mertens function—precisely when $M(p)/\sqrt{p}$ exceeds a threshold of roughly 0.14.

4.2 The natural conjecture and its counterexample

The computational evidence for $p \leq 50{,}000$ (zero exceptions among 5,129 primes) strongly suggested the conjecture:

Conjecture 4.1 (Disproved)**.** *For all primes $p \geq 11$: $M(p) < 0 \implies \Delta W(p) \leq 0$.*

This conjecture is **false**.

Observation 4.2 (Counterexample at $p = 92{,}173$)**.** At the prime $p = 92{,}173$:

$$M(92{,}173) = -2, \qquad \Delta W(92{,}173) = +3.56 \times 10^{-11}.$$

The Mertens function is strictly negative, yet the wobble decreases. This is confirmed by three independent computations:

(i) Naive double-precision floating point.

(ii) Long double (80-bit extended precision).

(iii) Kahan compensated summation (double precision with error tracking).

All three agree on the sign and magnitude to within rounding error. This is the *only* counterexample among 9,588 primes up to $p = 100{,}000$.

4.3 The $M = -2$ boundary

The counterexample lies at the shallowest possible negative Mertens value: $M(p) = -2$ (since $\mu(p) = -1$, each prime decreases M by exactly one). The normalized Mertens value $M/\sqrt{p} = -0.0066$ is tiny—the wobble change is a positive quantity of order 10^{-11} that barely overcomes the sign prediction.

4.4 The $M \leq -3$ class: zero counterexamples

Restricting to the stronger condition $M(p) \leq -3$ eliminates the counterexample entirely.

Observation 4.3. Among all 4,617 primes $p \leq 200{,}000$ with $M(p) \leq -3$:

$$\Delta W(p) \leq 0 \quad \text{in every case.}$$

Zero counterexamples. The tightest case has $|\Delta W| \approx 10^{-12}$, still safely negative.

Whether a threshold M_0 exists such that $M(p) \leq M_0$ guarantees $\Delta W(p) \leq 0$ for all primes remains open. The data is consistent with $M_0 = -3$, but the counterexample at $p = 92{,}173$ counsels caution—it appeared far beyond the range where the original conjecture seemed secure.

4.5 Rubinstein–Sarnak connection

The violation rate of $\sim$21% is reminiscent of the logarithmic densities in the Rubinstein–Sarnak theory of Chebyshev's bias [7]. In that theory, the proportion of integers where $\pi(x; 4, 3) > \pi(x; 4, 1)$ is determined by the zeros of Dirichlet L-functions. Our violation rate may be similarly governed by the distribution of the Mertens function, which depends on the zeros of $\zeta(s)$. The per-step wobble decomposition provides a new geometric observable through which to study this distribution.

5 Why Primes Are Special

The wobble–Mertens correlation is specific to primes. For composites $N \leq 50{,}000$, there are 97 instances where $\Delta W(N) > 0$ yet $M(N) < 0$. The mechanism behind this specificity is clear:

Proposition 5.1 (Formally verified)**.** *For a prime p, the integer p adds exactly $\varphi(p) = p - 1$ new fractions k/p ($k = 1, \ldots, p-1$) to $\mathcal{F}_{p-1}$. These are* equispaced *with gap $1/p$.*

Every new fraction k/p lands at the mediant of its enclosing Farey gap—the insertion is maximally uniform. For a composite $N = ab$, only the $\varphi(N) < N - 1$ coprime numerators appear, creating an irregular insertion pattern. This 100% grid fill is what allows the identities of Section 3 to yield clean formulas at prime arguments; at composites, the partial fill introduces additional error terms that break the sign control.

6 Formal Verification

All identities in Section 3 and their supporting lemmas have been formally verified in Lean 4 using the Mathlib library, with the Aristotle automated theorem prover [1] generating proof strategies. The formalization contains 60+ theorems with **zero** `sorry` declarations.

6.1 Key verified results

1. **Farey involution**: $\sigma(a/b) = (b - a)/b$ is a bijection on $\mathcal{F}_N$.
2. **Involution displacement**: $D(f) + D(\sigma(f)) = -1$.
3. **Master Involution Principle**: $\sum D \cdot g = -\frac{1}{2} \sum g$ for symmetric g (Theorem 3.3).
4. **Bridge Identity** (Theorem 3.1).

5. **Displacement–Cosine Identity** (Theorem 3.4).
6. **Fractional Parts Sum** (Theorem 3.6).
7. **Fractional parts involution**: $\{pf\} + \{p(1-f)\} = 1$ for $0 < f < 1$.
8. **Farey symmetry**: $\sum_{f \in \mathcal{F}_N} f = |\mathcal{F}_N|/2$.
9. **Wobble decomposition**: $W = S_2 - 2R/n + J(n)$.
10. **New fractions sum**: $\sum_{k=1}^{p-1} (k/p)^2 = (p-1)(2p-1)/(6p)$.
11. **Ramanujan sum at coprime argument**: $c_q(m) = \mu(q)$ when $\gcd(q, m) = 1$.
12. **Prime grid fill**: primes contribute exactly $p - 1$ new fractions.
13. **Farey cardinality recurrence**: $|\mathcal{F}_N| = |\mathcal{F}_{N-1}| + \varphi(N)$.
14. **Insertion orthogonality**: $\sum D(k/p) \cos(2\pi k/p) = 0$.

6.2 The role of Aristotle

The Aristotle theorem prover [1] combines LLM-guided proof search with Lean 4 tactic automation. It was essential for lemmas requiring non-trivial tactic chains (`ring_nf`, `grind`, `omega`, `norm_num`), particularly the Ramanujan sum evaluations and partition identities where manual proof construction would have been prohibitive.

7 Applications and Connections

7.1 A new bridge between research communities

The bridge identity (Theorem 3.1) establishes a literal equality between an object from Farey geometry (a Fourier coefficient of the sequence at prime frequency) and an object from multiplicative number theory (the Mertens function). Results about $M(p)$ immediately yield consequences for Farey discrepancy, and vice versa. While the ingredients are classical, the particular synthesis—evaluating the Fourier transform of the Farey sequence at prime frequency and recognizing the Mertens function—appears new.

7.2 Erdős–Turán at prime frequencies

The Erdős–Turán inequality [3] bounds discrepancy in terms of exponential sums. At prime frequency $m = p > N$, our bridge identity gives the *exact* value $M(p) + 2$ for the relevant exponential sum—previously only bounded. The sharp form due to Shu and Wang [8] suggests that exact evaluations at multiple frequencies could yield improved discrepancy bounds.

7.3 García's local Farey estimates

García [5] recently obtained new analytical formulas for the rank of Farey fractions, providing local discrepancy estimates. Our displacement–cosine identity (Theorem 3.4) gives a complementary global perspective: the Mertens function controls the Fourier-weighted *sum* of all rank discrepancies at prime frequency, while García's formulas control individual rank values.

7.4 Rubinstein–Sarnak: violation rate as geometric observable

The $\sim$21% violation rate provides a new geometric observable—the density of primes that improve Farey uniformity—whose asymptotic value may be expressible through the Rubinstein–Sarnak framework [7]. If so, this would connect the zeros of $\zeta(s)$ to the per-step geometry of Farey sequences in a previously unknown way.

7.5 Franel–Landau: per-step refinement

The telescoping identity $W(N) = W(1) - \sum_{k=2}^{N} \Delta W(k)$ expresses the cumulative discrepancy as a sum of individual contributions. The wobble–Mertens phenomenon constrains the sign structure of this sum: when $M(p)$ is sufficiently negative, the prime p's contribution is non-negative (it does not help decrease the wobble). The "healing" contributions that drive $W(N) \to 0$ come predominantly from primes during positive Mertens excursions and from composites.

8 Open Questions

1. **Threshold.** Is there a value $M_0 \leq -3$ such that $M(p) \leq M_0$ implies $\Delta W(p) \leq 0$ for all primes p? The $M \leq -3$ class has zero counterexamples among 4,617 primes through $p = 200{,}000$.

2. **Quadrature defect.** The gap between the identities of Section 3 and a full proof lies in bounding the "R^2 quadrature defect"—the antisymmetric component $\sum D(f)(\{pf\} - 1/2)$. Can this be bounded by Cauchy–Schwarz together with RH-conditional discrepancy estimates?

3. **Other discrepancy measures.** Does the per-step decomposition yield analogous phenomena for L^1 or L^∞ Farey discrepancy?

4. **Exact violation density.** Does the proportion of primes with $\Delta W(p) > 0$ converge to a constant? Is it expressible through the Rubinstein–Sarnak density?

5. **Applications.** The equispaced insertion property of primes is reminiscent of quasi-Monte Carlo point sets. Can the per-step decomposition inform the design of low-discrepancy sequences or find applications in zero-knowledge proof systems (where Farey fractions and modular arithmetic already appear)?

Acknowledgments

The formal verification was carried out using the Aristotle automated theorem prover [1]. The compact cross-term formula (Theorem 3.8) was discovered during a collaborative session with a GPT-based agent. Computation was performed in C using the Farey next-term generator, with results verified by Python exact rational arithmetic for small primes. We thank the Mathlib community for maintaining the Lean 4 library that made the formalization possible.

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